

ENGI 9839: Software Verification and Validation

Project Topic Proposal

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Proposed Topic:

Comparing Automated Dynamic UI Verification to Static Walkthroughs in Asynchronous Systems

Research Question:

How effective is automated dynamic validation (using **Playwright E2E testing**) at detecting UI state anomalies in asynchronous, real-time dashboards compared to manual static walkthroughs?

Motivation

Per the project guidelines, this topic meaningfully extends the course's coverage of dynamic and static validation. The foundational literature establishes the differences between manual static reviews and dynamic testing, but largely assumes synchronous execution. This project extends that foundation by investigating the unique challenges of verifying non-deterministic, asynchronous software behavior, specifically WebSocket-driven UI state mutations, and evaluating modern automated solutions.

Methodology Sketch

We will conduct a case study on an **open-source observability platform (Uptime Kuma)** to evaluate these implementation-phase techniques.

- 1. Static Validation Baseline:** We will conduct manual static walkthroughs of the dashboard's real-time state updates to establish a baseline for defect detection and time expenditure.
- 2. Dynamic Validation Prototype:** We will engineer an automated testing pipeline using Playwright to dynamically verify those same asynchronous UI components.
- 3. Evaluation:** The final report will provide a comparative analysis of the defect detection rate, execution speed, and overall test adequacy of the automated dynamic approach versus the manual static approach in a real-time environment.